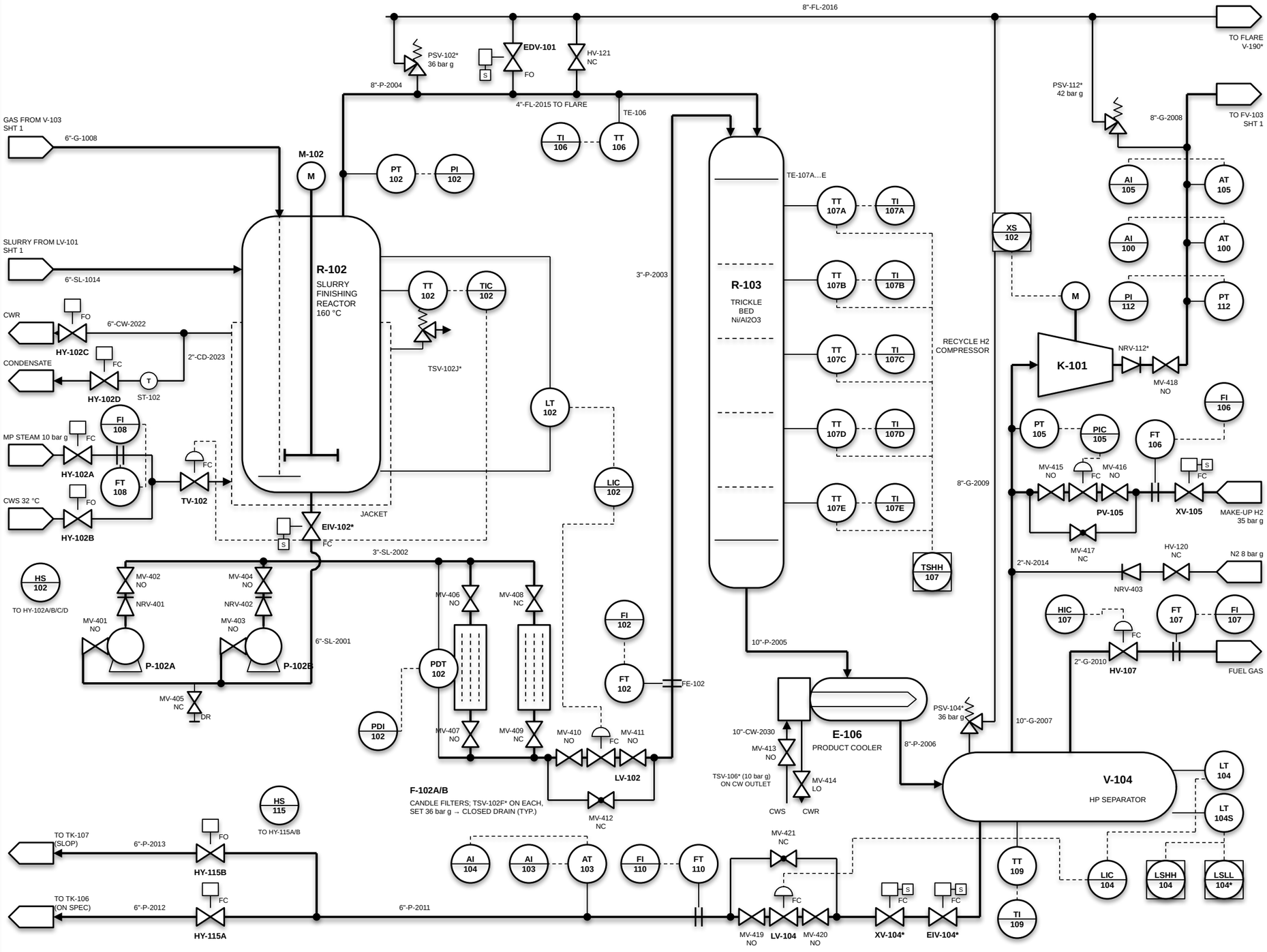


INTERLOCK SCHEDULE (SIS-100)

from model.js: tripConditions(), TRIPDLY, bmsTrip(), bmsCmd(); legend on sheet 1

TRIP	CAUSE / SENSOR	SETTING	DELAY	ACTION
TSHH-100	R-101 temperature, TE-100 / TT-100	> 220 °C	5 s	UNIT TRIP
TSHH-107	R-103 bed, hottest of TE-107A…E	> 220 °C	5 s	UNIT TRIP
PSHH-101	R-101 pressure, PT-101	> 33.5 bar g	3 s	UNIT TRIP
XS-101	M-101 stopped while XV-101 open (run signal)	motor stopped	10 s	UNIT TRIP
XS-102	K-101 stopped while XV-101 open (run signal)	motor stopped	10 s	UNIT TRIP
LSHH-104	V-104 level, LT-104S (liquid toward K-101)	> 90 %	5 s	STOP K-101 ONLY (not latched)
HS-100	ESD pushbutton (operator)	manual	—	UNIT TRIP + EDV-101 OPENS
LSLL-104*	V-104 level, LT-104S (gas blow-by)	5 % (assumed)	TBC	XV-104* SHUTS (SL-R6, HOLD)
LSLL-109*	D-101 level (own element, HOLD)	10 % (assumed)	TBC	UNIT TRIP (SL-R3, HOLD)

UNIT TRIP ACTIONS (bmsTrip, latched, first-out recorded):

- XV-101 benzene and XV-105 make-up H2 close (de-energise).
- K-101 stops.
- FIC-101, FIC-103, PIC-105 to MAN, output 0 %: FV-101, FV-103, PV-105 close behind the SIS valves.
- HS-101 → COOL: HY-101A/B open, HY-101C/D shut (MP steam off); FV-104 forced 100 % (full pump-around through E-101).
- HS-102 → COOL: HY-102B/C open, HY-102A/D shut; TV-102 100 %.
- ESD (HS-100) only: EDV-101 opens to flare.
- Recommended (SL-R7, HOLD): stop P-100A/B on unit trip.

RESET (bmsCmd reset):

- Blocked while a process trip condition is active or TI-100 > 210 °C.
- After reset H2 and benzene stay isolated until admitted again.

PERMISSIVES (purgePermissives, bmsCmd h2 / feed):

- XV-105 (H2) may open only with loop O2 AI-100 < 1 %.
- XV-101 (benzene) may open only with ALL of: SIS reset, no trip latched · XV-105 open and AI-105 ≥ 80 % · LIC-101 ≥ 40 % · M-101 running · K-101 running and FIC-103 ≥ 15 000 Nm³/h · TI-100 ≥ 140 °C (light-off) · E-103 on CW (HV-113 open).

FAIL POSITIONS ON INSTRUMENT AIR FAILURE (step() FAIL map):

- FC: FV-101, FV-103, PV-105, LV-101, TV-102, LV-102, LV-104, HV-107.
- FO: FV-104 (full flow through E-101), LV-109 (BFW), PV-110 (LP steam).
- SIS valves XV-101, XV-105, XV-104*, EIV-1nn* FC; EDV-101 FO (assumed).
- HY valves: fail to the COOL line-up (HY-101A/B, HY-102B/C FO).

NOTES:

1 DESIGN CASE (OTS model at design): R-102 160 °C / 29.60 bar g, level 55 %, jacket on MP steam 2.33 t/h (HS-102 HEAT); liquid to R-103 20.9 t/h, F-102 ΔP 0.26 bar (alarm 0.8); R-103 bed 158.5-158.6 °C; V-104 29.0 bar g / 44.9 °C, level 50 %; product 37.5 t/h, benzene 2.0 ppm, MCP 60 ppm; recycle 46,000 Nm³/h at 92.1 % H2; make-up 31,974 Nm³/h; purge 2,017 Nm³/h (HIC-107 25 %); K-101 discharge 32.0 bar g.

2 Loop pressure profile (model): K-101 32.0 → R-101 30.0 → R-102 29.6 → R-103 29.25 → V-104 29.0 bar g. K-101 discharge 57 °C by isentropic compression (η 0.70 assumed) vs 60 °C in the model.

3 PSV-102*, PSV-104* set 36 bar g = loop design pressure (assumed); PSV-112* set 42 bar g (K-101 deadhead 35 × 1.154 = 40.4 bar g); TSV-102J*, TSV-106* 10 bar g; TSV-102F* 36 bar g. Relief loads and flare header size: HOLD. V-190* flare knock-out drum is off this drawing.

4 Line sizes preliminary (see sheet 1 note 2). Instrument and SIS conventions, MV / NRV numbering and the * mark: see sheet 1 legend and notes.

5 PT-105 is on the V-104 gas outlet to K-101 (same pressure as V-104). TT-109 is on V-104. TE-107A…E are shared by TI-107A…E and TSHH-107 (separate SIS elements TE-107S: SRS recommendation, HOLD).

6 F-102A in service (inlet / outlet MVs open), F-102B on stand-by (shut, drained). Change-over: open B outlet, then B inlet, shut A inlet, then A outlet; A is drained under N2 and kept wet (spent Raney-Ni is pyrophoric, CAT-2).

7 HS-102 (DCS selector) HEAT: HY-102A (MP steam) and HY-102D (condensate) open; COOL: HY-102B (CWS) and HY-102C (CWR) open. TIC-102 acts reverse in HEAT and direct in COOL (model). HY-102A-D and ST-102 are new tags of this drawing (HOLD).

8 HS-115 (DCS) selects HY-115A to TK-106 (on spec) or HY-115B to TK-107 (slop). New tags; fail positions TBC (drawn: TK-107 open on air failure).

9 K-101 minimum-flow / anti-surge protection and seal system not drawn (HOLD: vendor package). The OTS alarms "K-101 LOFL" when FV-103 is shut with K-101 running.

10 N2 (HV-120) purges the loop below 1 % O2 before XV-105 opens; HV-121 vents to flare. EDV-101 is opened only by the ESD (HS-100) or by hand from the SIS page.

11 TI-120 (CW supply temperature, page indicator) is measured on the CW header OSBL and is not drawn; TI-107 (hottest bed temperature) is a DCS calculation from TT-107A…E.

REV	DESCRIPTION	DATE
—	SCREEN P&ID IN OTS PAGE: CURRENT, USED IN DECKS	
0	ISSUED FOR TEACHING · ALT. VERSION, EPC STD	2026-10-26

CYCLOHEXANE REACTOR VER 1 — OTS TEACHING UNIT

P&ID — CYCLOHEXANE REACTOR VER 1 · ALTERNATIVE VERSION
SHEET 2: R-102 · R-103 · V-104 · K-101 · INTERLOCKS

DRAWN	CHK'D	APP'D	DATE
Apinan Soottitantawat	—	—	2026-10-26

DWG NO.	REV	SCALE	SHEET
CHX-V1-PID-001	0	N.T.S.	2 OF 2